

Synoptic CB: Porewater Nutrients

May 2023 - DO NOT RUN REQUIRES SAMPLE RERUNS

2025-12-22

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##Run Information

```
cat("Run Information: Isabelle Van Benschoten") #lets you know what section you're in
```

Run Information: Isabelle Van Benschoten

```
#set the run date & user name
run_date <- "20230603/20230111"
sample_year <- 2023
sample_month <- 05
user <- "Stephanie Wilson"
```

```

#identify the files you want to read in
#read in as a list to accommodate multiple runs in a month
files <- c("Raw Data/COMPASS_Synoptic_Nutrients_clean_2023_May.csv")
file_stds <- ("Raw Data/COMPASS_Synoptic_Nutrients_df_all_2023.csv")

# Define the file path for QAQC log file - NO Need to change just check year
file_path <- "Raw Data/SEAL_COMPASS_Synoptic_QAQC_Log_2023.csv"
final_path <- "Processed Data/COMPASS_Synoptic_Nutrients_May2023.xlsx"

#record any notes about the run or anything other info here:
run_notes <- "Multiple months of data were mixed together so they were parsed
and cleaned prior to read in.
Samples ran on 01/11/24 were not included because of boor standard curve.
Autospikes for PO4 not included in this run.
Sample TEMPEST_20230513_CON_C6 was input twice.
NH3 and PO4 have low number of duplicates due to the bad standard curve run"

#duplicate sample names to be changed
#list the sample IDs that are messed up and create a list
#with run number as well so that we can change them below
# wrong_names <- c("GCW_202304_TR_LysC_45cm", "GCW_202304_TR_LysA_20cm_8",
#                  "GWI_202304_UP_LysA_20cm", "GWI_202304_UP_LysA_20cm")
# wrong_nums <- c(20, 16, 46, 44)
# correct_names <- c("GCW_202304_TR_LysB_45cm", "GCW_202304_TR_LysA_20cm",
#                    "GWI_202304_UP_LysA_10cm", "GWI_202304_UP_LysA_10cm")

#can't determine from metadata - for now unsure
# remove_names <- c("GCW_202304_TR_LysA_20cm", "GCW_202304_TR_LysA_20cm",
#                  "GCW_202304_TR_LysB_20cm_13", "GCW_202304_TR_LysB_20cm_13")
# #couldn't tell which one this is from the metadata, no A_10cm which is what we thought
#marked on the sheet, need to check sample vials in freezer
#to see if we have a A_10cm from GCW_TR to be sure
#remove_nums <- c(15, 13, 21, 19 )

#Set up file path for metadata
#downloaded metadata csv - downloaded from Google drive as csv for this year
#https://docs.google.com/spreadsheets/d/1HCANO_q6y17x0RUXVzID09hVal-RfwWc/edit?usp=sharing&ouid=10899
Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2023.csv"

```

```
##Setup
```

```
##Read in metadata and create similar sample IDs for matching to samples
```

0.1 Import Data & Clean

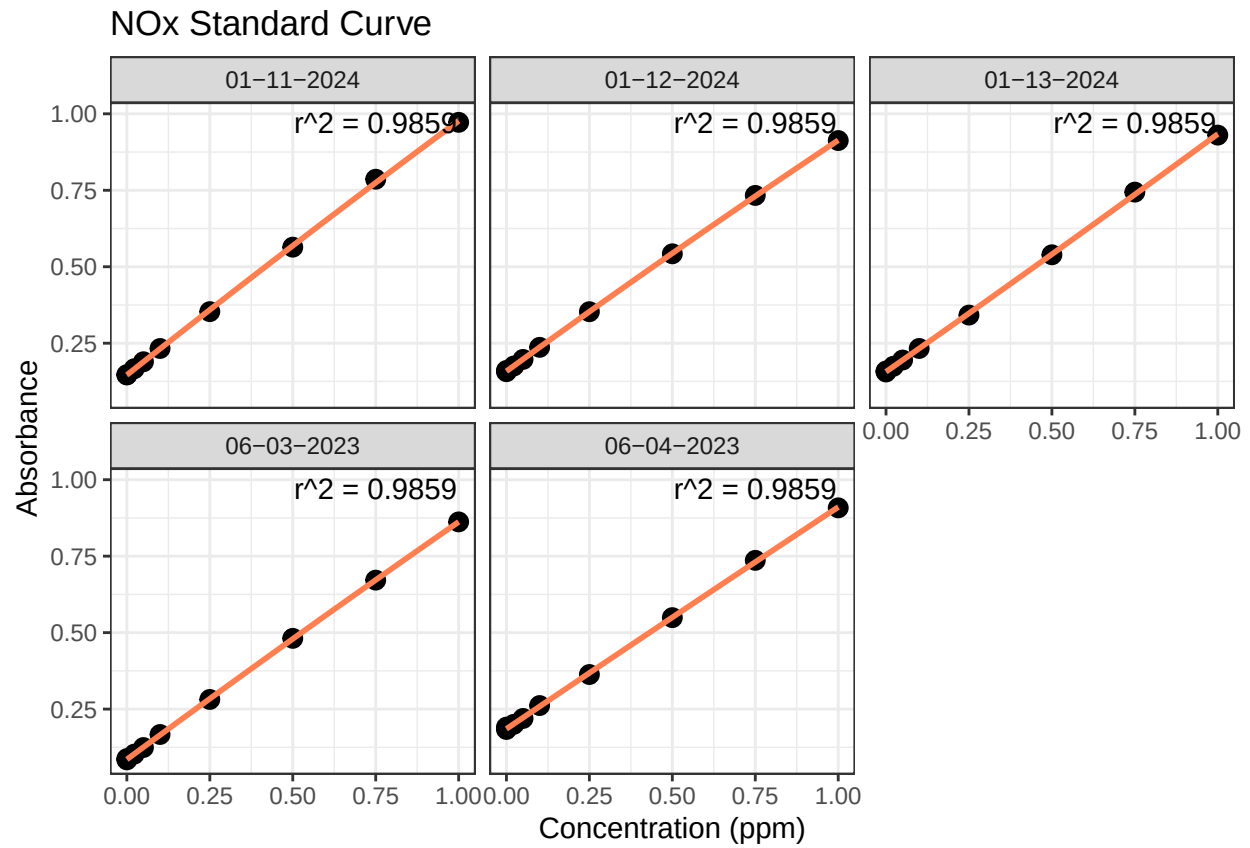
0.2 Assessing standard Curves

```
#Pull out standards data
```

```
## Assess Standard Curves
```

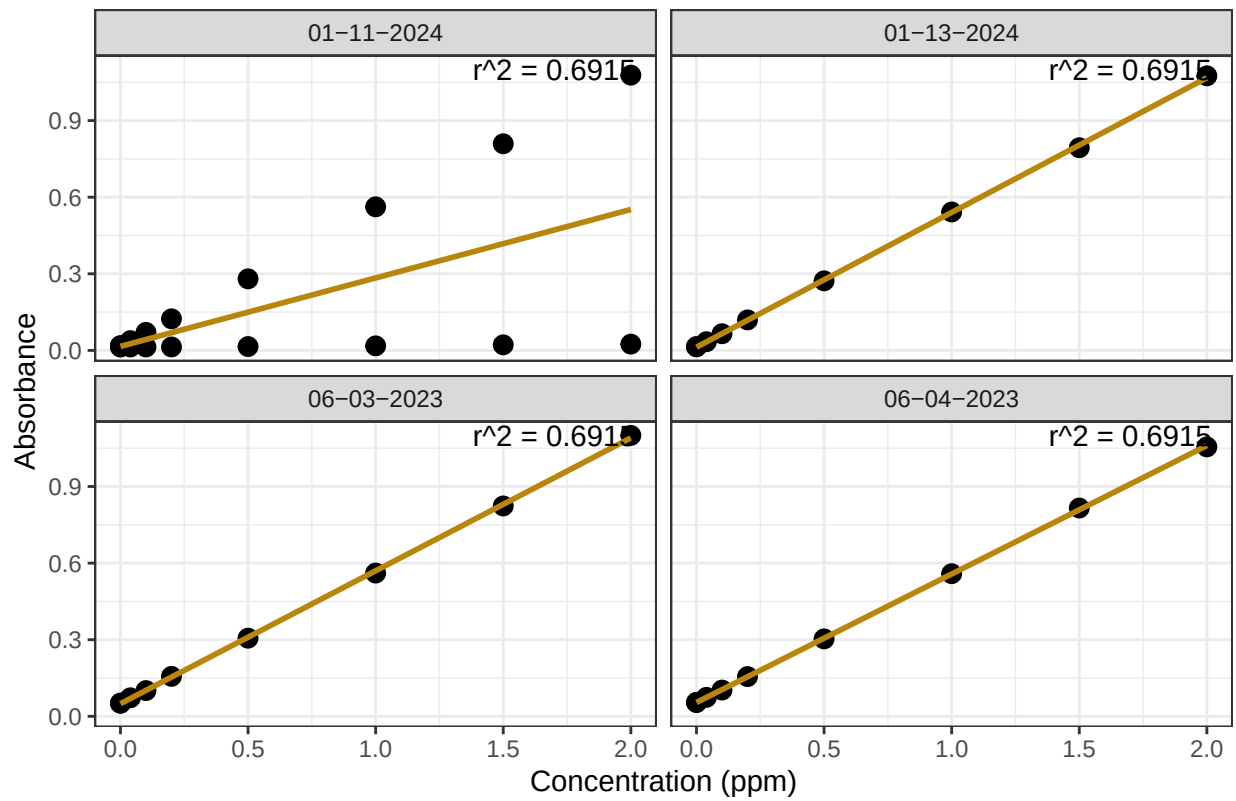
```
#Plot standards data
```

```
## Assess Standard Curves
```



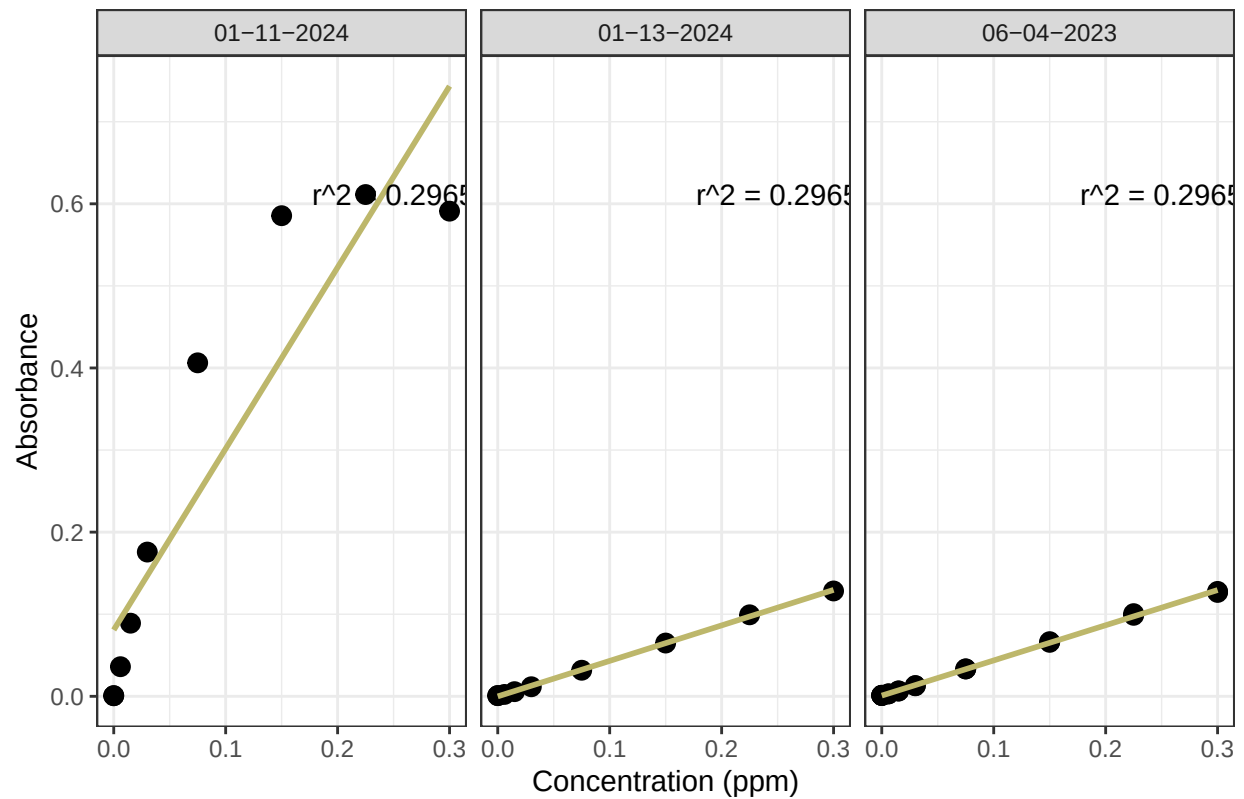
```
## 'geom_smooth()' using formula = 'y ~ x'
```

NH3 Standard Curve



```
## 'geom_smooth()' using formula = 'y ~ x'
```

PO4 Standard Curve



```
## [1] "NOx Curve r2 is below cutoff! - REASSESS"
```

```
## [1] "NH3 Curve r2 is below cutoff! - REASSESS"
```

```
## [1] "PO4 Curve r2 is below cutoff! - REASSESS"
```

```
## [1] "QAQC log file exists and has been read into the code."
```

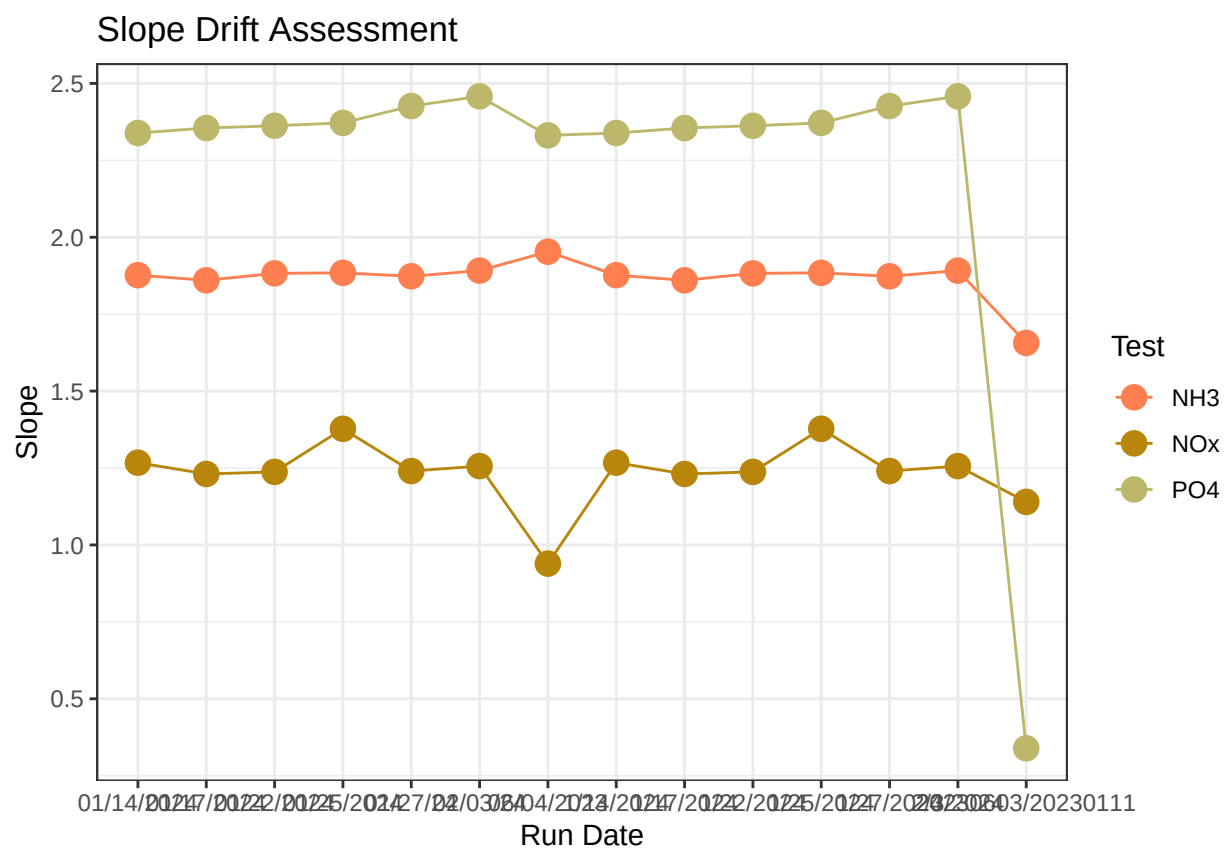


Table 1: Average Slope by Analyte

Test	avg_slope
NH3	1.868
NOx	1.236
PO4	2.235

```
cat("Removing Data from Bad Curve on 01/11/24")
```

```
## Removing Data from Bad Curve on 01/11/24
```

```
df_all_may <- df_all_stds %>%  
  filter(!str_detect(Sample_Name, "202304"))
```

```
df_all <- df_all_may %>%  
  mutate(Run_Time = mdy_hm(Run_Time),  
         Run_Date = as_date(Run_Time)) %>%  
  filter(!(Run_Date == as_date("2024-01-11") & Test %in% c("o-PHOS 0.3", "Ammonia 2")))
```

0.3 Dilution Corrections - ensure the latest dilution is kept

```
## Dilution Corrections
```

```
## Duplicated samples: TEMPEST_20230513_CON_C6
```

```
## Duplicated samples with NO dilution present (possible input error): TEMPEST_20230513_CON_C6
```

0.4 Temporary Chunk

0.5 Performance Check

```
## [1] "NOx pe Check has a % Difference <10% - PROCEED"
```

```
## Run mean = 0.712214
```

```
## Expected = 0.706
```

```
## [1] "NH3 pe Check has a % Difference >10% - REASSESS"
```

```
## Run mean = 1.357356
```

```
## Expected = 0.948
```

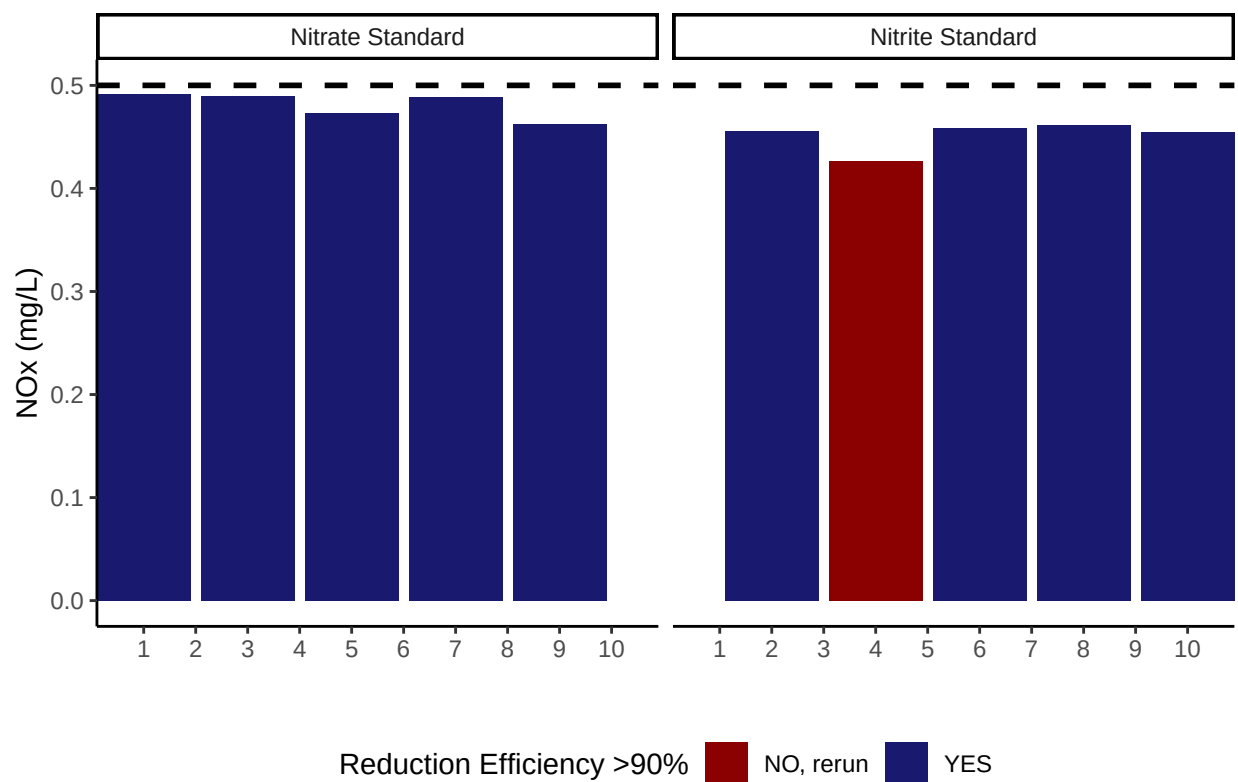
```
## [1] "P04 pe Check has a % Difference <10% - PROCEED"
```

```
## Run mean = 0.8833125
```

```
## Expected = 0.818
```

0.6 Check NOx Reduction Efficiency

```
## Assess Reduction Efficiency
```

```
## [1] "Mean NOx Reduction Efficiency <95% - REASSESS"
```

```
## [1] 93.24452
```

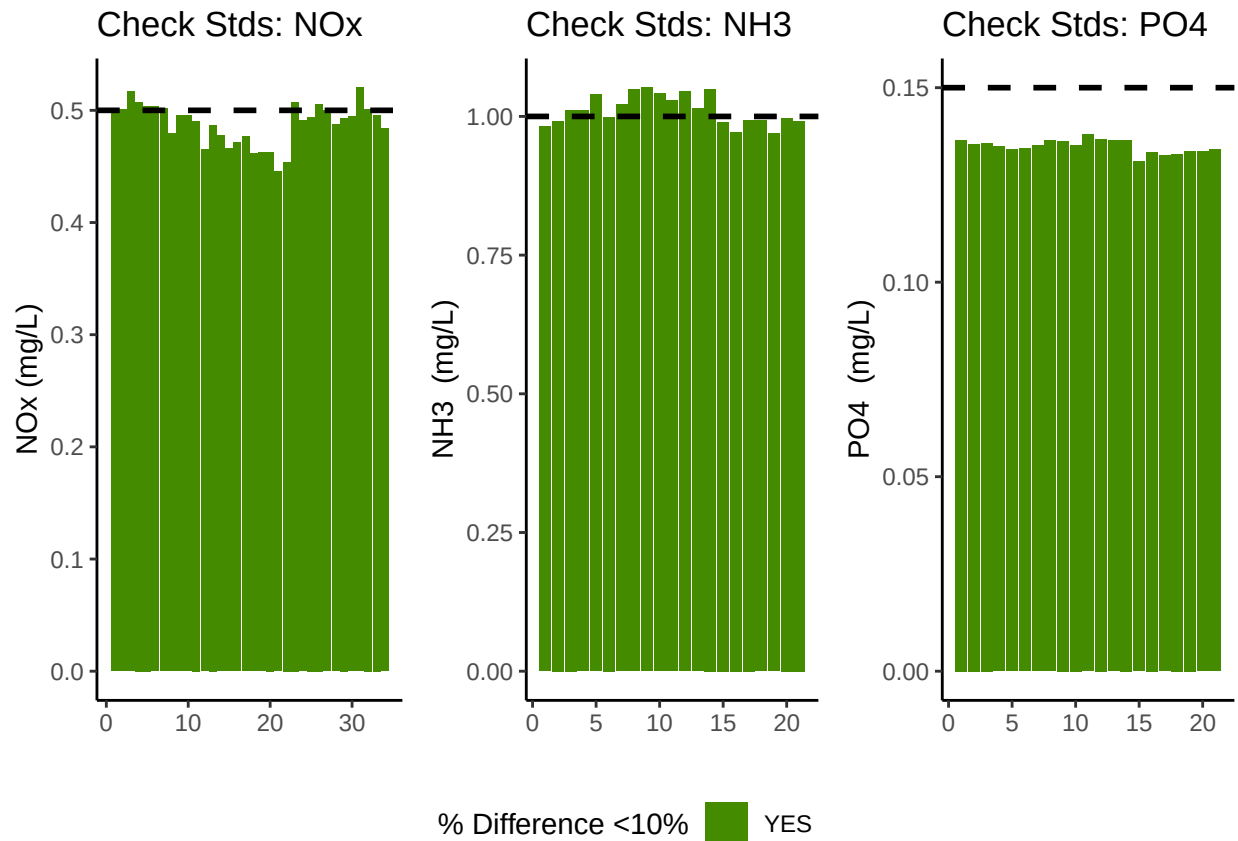
0.7 Analyze the Check Standards

```
## Analyze Check Standards
```

```
## [1] "NOx Check Standard RSD within Range - PROCEED"
```

```
## [1] "NH3 Check Standard RSD within Range - PROCEED"
```

```
## [1] "PO4 Check Standard RSD within Range - PROCEED"
```



```
## [1] ">60% of NOx Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">60% of NH3 Check Standards are within range of expected concentration - PROCEED"
```

```
## [1] ">60% of PO4 Check Standards are within range of expected concentration - PROCEED"
```

0.8 Analyze Blanks

```
## Assess Blanks
```

```
## [1] ">60% of NOx Blank concentrations are lower than the lower 25% quartile of samples - PROCEED"
```

```
## [1] ">60% of NH3 Blank concentrations are lower than the lower 25% quartile of samples - PROCEED"
```

```
## [1] ">60% of PO4 Blank concentrations are lower than the lower 25% quartile of samples- PROCEED"
```

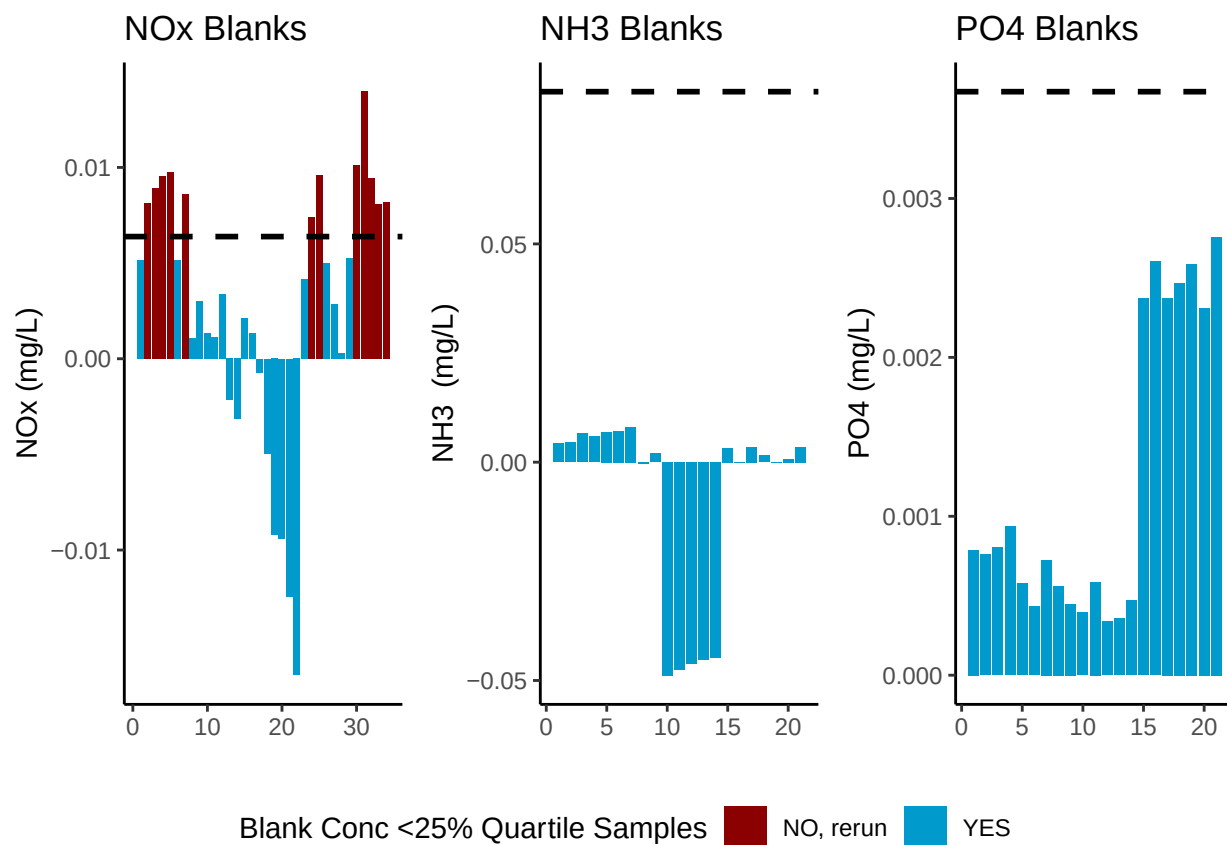


Table 2: Mean Concentration of Blanks

Test	Blank_Mean_Conc
NOx	0.0028
NH3	-0.0083
PO4	0.0012

0.9 Analyze Duplicates

```
## Analyze Duplicates

## Warning: Returning more (or less) than 1 row per 'summarise()' group was deprecated in
## dplyr 1.1.0.
## i Please use 'reframe()' instead.
## i When switching from 'summarise()' to 'reframe()', remember that 'reframe()'
## always returns an ungrouped data frame and adjust accordingly.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

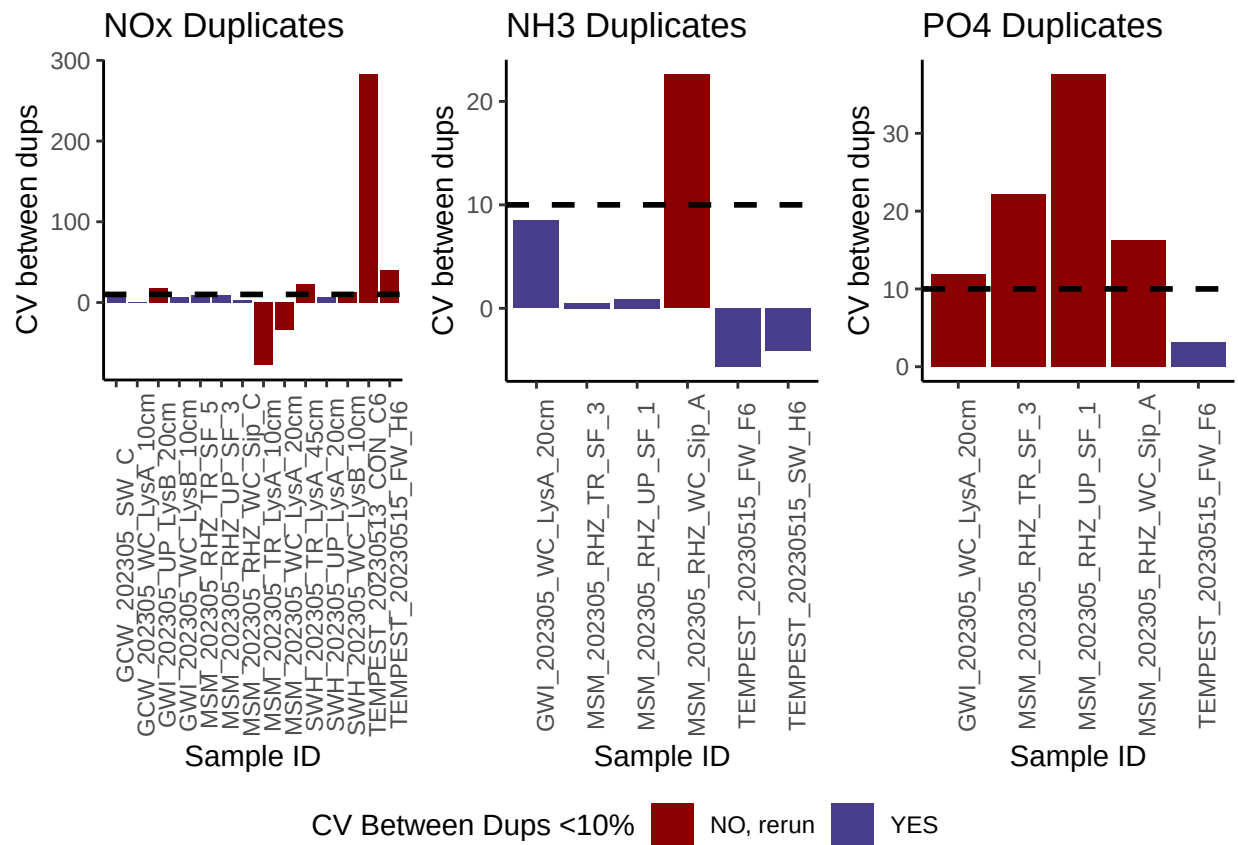
## # A tibble: 6 x 3
##   Pair_ID      Type      n_obs
##   <chr>      <chr>    <int>
## 1 0.15 mg/L    Original     21
## 2 Abs_Chk_10ppt Original      3
## 3 Abs_Chk_20ppt Original      3
## 4 Blank      Duplicate      7
## 5 Blank      Original     21
## 6 peChk_NH3_PO4 Original      2

## [1] "<60% of NOx Duplicates have a CV <10% - REASSESS"

## [1] ">60% of NH3 Duplicates have a CV <10% - PROCEED"

## [1] "<60% of PO4 Duplicates have a CV <10% - REASSESS"

## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

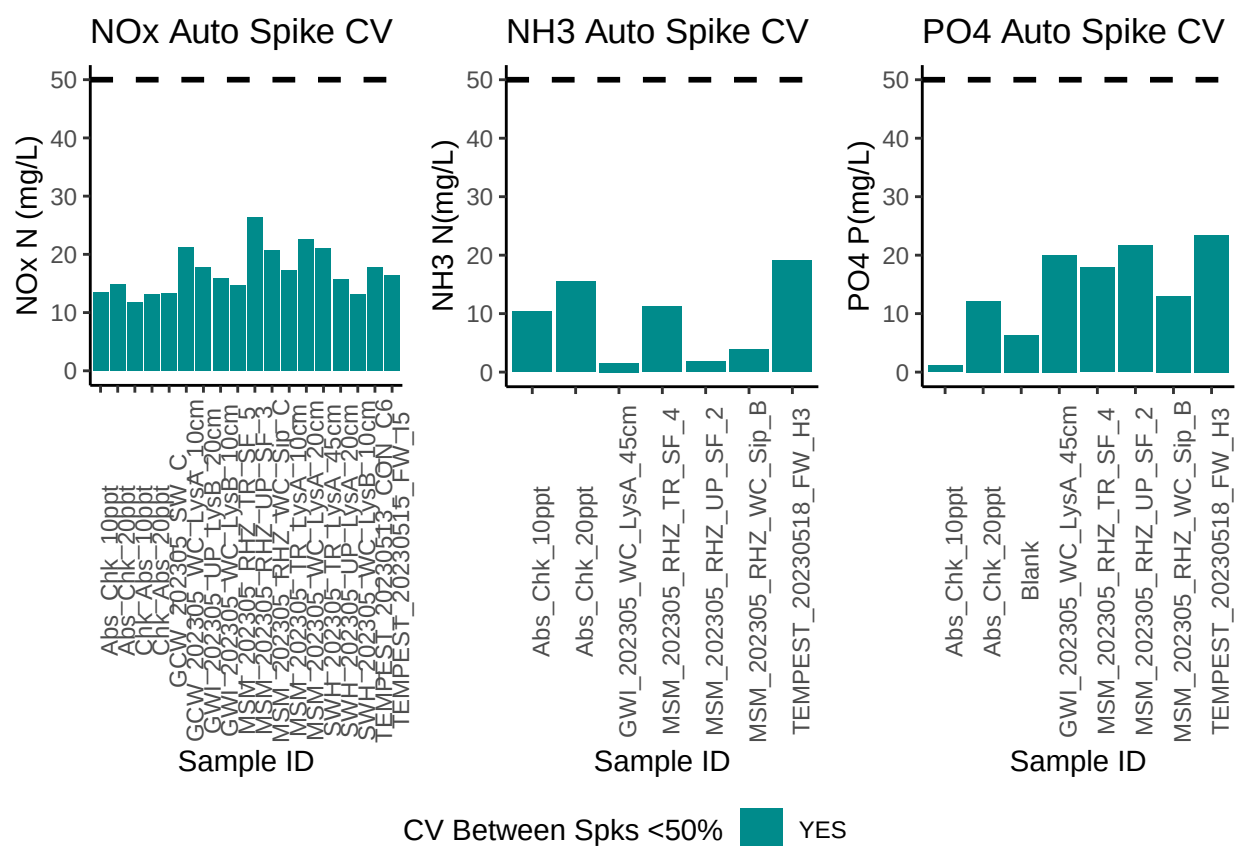


0.10 Spikes

```
## [1] ">60% of Spikes have a CV <50% - PROCEED"
```

```
## [1] ">60% of Spikes have a CV <50% - PROCEED"
```

```
## [1] ">60% of Spikes have a CV <50% - PROCEED"
```



0.11 Matrix Effects

```
## [1] "NO NOx Matrix Effect, PROCEED"
```

```
## [1] "NO NH3 Matrix Effect, PROCEED"
```

```
## [1] "NO PO4 Matrix Effect, PROCEED"
```

0.12 Unit Converted Data Column Added (mg/L to uM)

0.13 Sample Flagging - Within range of standard curve

```
## Sample Flagging
```

0.14 Pull out sample id information

```
## Sample Processing
```

```
## Warning: There was 1 warning in 'mutate()'.  
## i In argument: 'Run_Time = lubridate::mdy_hm(Run_Time)'.  
## Caused by warning:  
## ! All formats failed to parse. No formats found.
```

```
## Warning: Expected 5 pieces. Missing pieces filled with 'NA' in 17 rows [19, 20, 21, 46,  
## 47, 48, 83, 84, 108, 109, 110, 125, 126, 127, 142, 143, 144].
```

0.15 Pulling Out Rhizon Samples

0.16 Check to see if samples run match metadata & merge info

```
## Check Sample IDs with Metadata
```

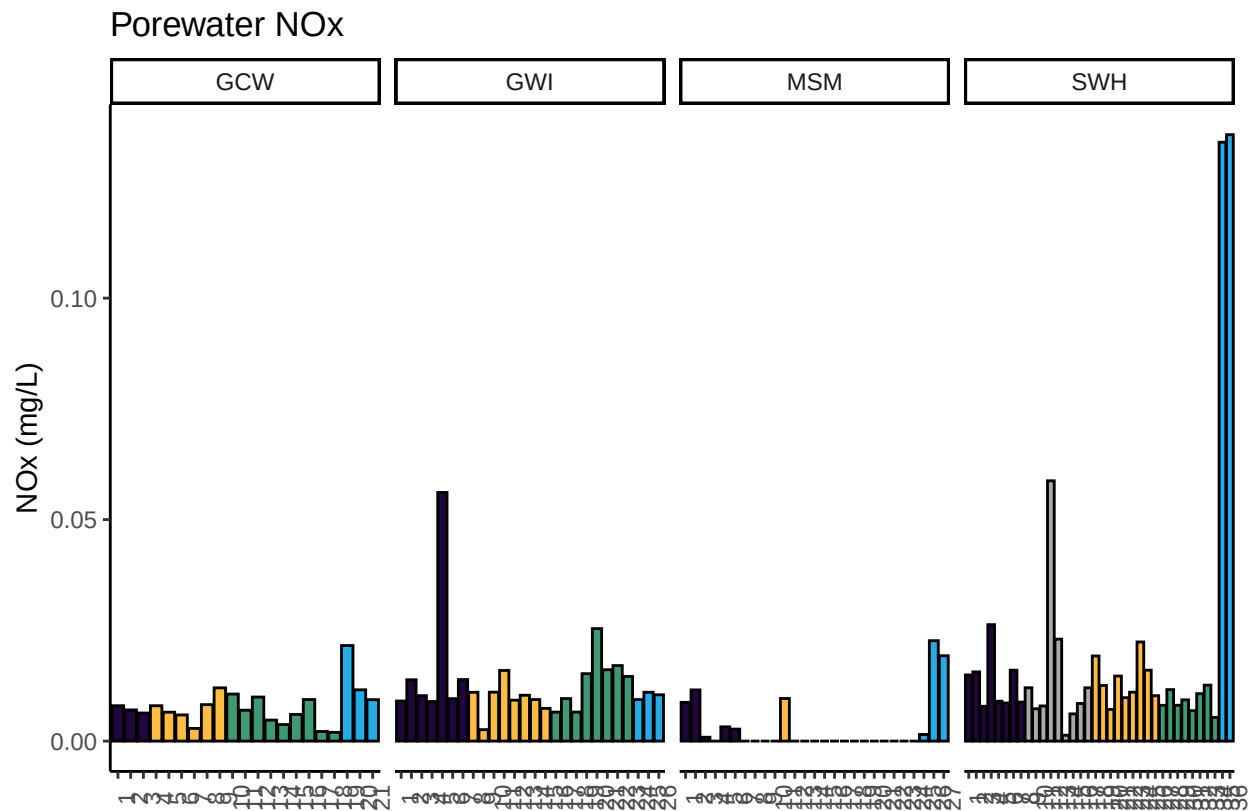
```
## Some sample IDs are missing from metadata.
```

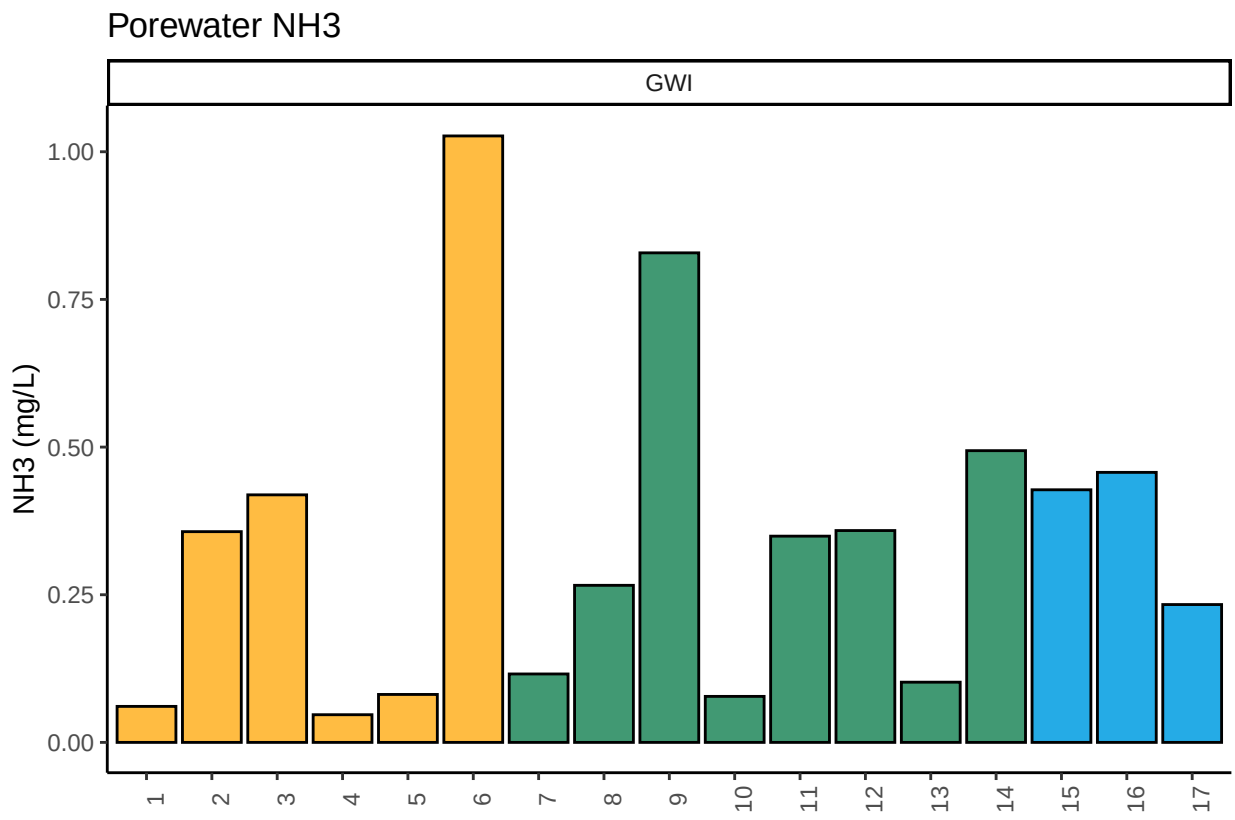
```
## [1] "GCW_202305_UP_LysA_20cm" "GCW_202305_UP_LysB_10cm"  
## [3] "GCW_202305_UP_LysB_20cm" "MSM_202305_UP_LysC_45cm"  
## [5] "MSM_202305_TR_LysC_45cm" "MSM_202305_WC_LysA_45cm"  
## [7] "SWH_202305_UPCON_LysA_10cm" "SWH_202305_UPCON_LysA_20cm"  
## [9] "SWH_202305_UPCON_LysA_45cm" "SWH_202305_UPCON_LysB_10cm"  
## [11] "SWH_202305_UPCON_LysB_20cm" "SWH_202305_UPCON_LysB_45cm"  
## [13] "SWH_202305_UPCON_LysC_20cm" "SWH_202305_UPCON_LysC_45cm"  
## [15] "SWH_202305_UP_LysB_10cm" "SWH_202305_WC_LysA_10cm"  
## [17] "SWH_202305_WC_LysA_20cm" "SWH_202305_WC_LysA_45cm"  
## [19] "SWH_202305_WC_LysB_10cm" "SWH_202305_WC_LysB_20cm"  
## [21] "GWI_202305_TR_LysB_45cm"
```

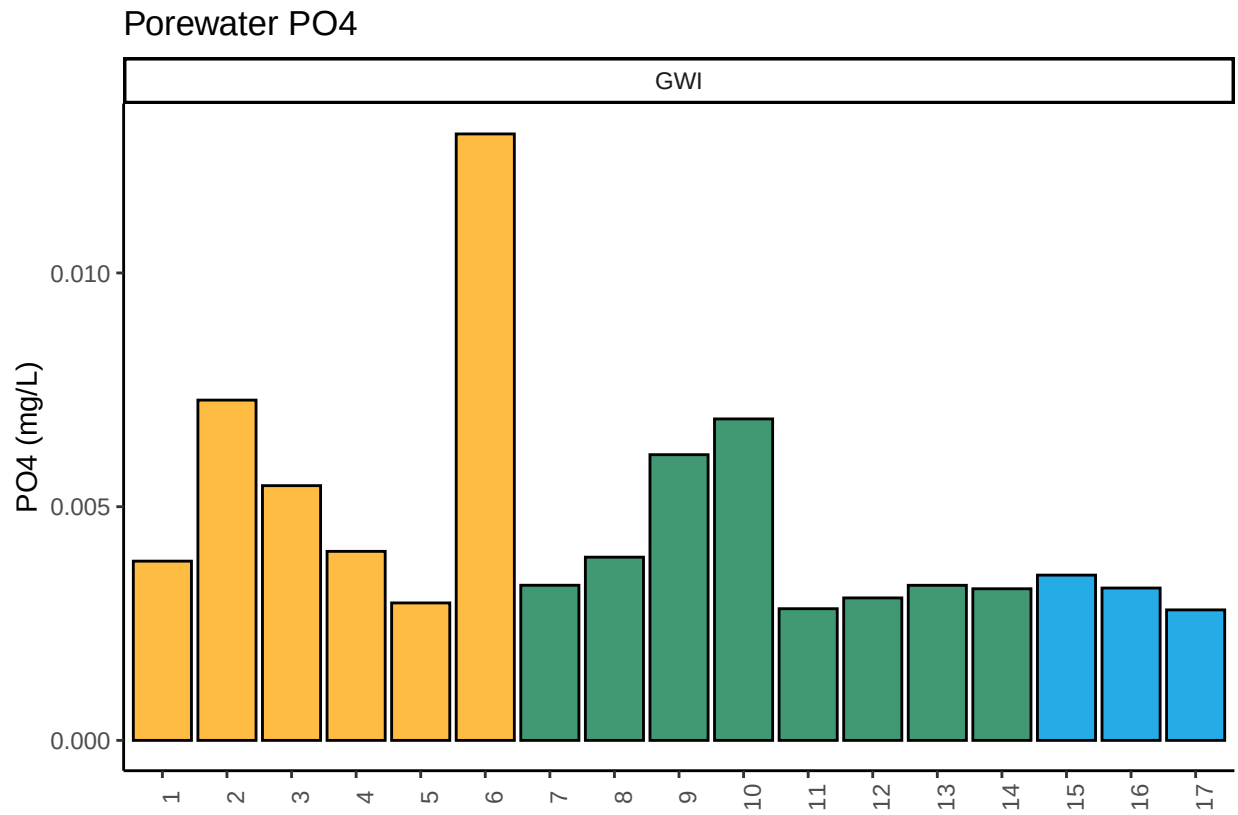
```
## Warning: Expected 5 pieces. Missing pieces filled with 'NA' in 17 rows [19, 20, 21, 46,  
## 47, 48, 83, 84, 108, 109, 110, 125, 126, 127, 142, 143, 144].
```

0.17 Visualize Data

Visualize Data







0.18 Export Processed Data

#end